Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 9: The Other Normal Form

Conjunctive Normal Form, Satisfiability, and SAT

Lehman, Leighton & Meyer, *MCS* Sections 3.3.2, 3.4.1, and 3.5

Name: _____ Date: _____

Where we are

Wednesday you built a formula from the **true** rows of a table and got an OR of ANDs. Today you do the same thing from the **false** rows and get an AND of ORs. Then we see why that second form is the one the world actually uses.
This closes our logic unit. Predicate logic and sets begin next week.

By the end of today you should be able to

1. Read a conjunctive normal form off a truth table.
2. Distinguish valid, satisfiable, and unsatisfiable.
3. Say what the SAT problem is and why it matters.

1. The mirror-image recipe

Conjunctive normal form

An **OR-term** (or **clause**) is a disjunction of literals: $P \vee (\sim Q)$.
A **conjunctive normal form (CNF)** is a conjunction of clauses — an **AND of ORs**.

Reading a CNF off a truth table

1. Look at each row where the formula is **false**.
2. For that row, write a clause: each variable *negated* if it is T in that row, *plain* if it is F. **(Opposite of the DNF rule.)**
3. AND all those clauses together.

Why it works. Each clause is false in exactly one row — the row it came from — and true everywhere else. ANDing them gives a formula false in precisely the rows you selected.

Keep the two recipes straight

	Built from	Shape	A variable that is T in that row
DNF	**true** rows	OR of ANDs	appears plain
CNF	**false** rows	AND of ORs	appears negated

Everything flips. If you remember one recipe you can reconstruct the other.

2. Worked together: $P \Rightarrow Q$ again

P	Q	$P \Rightarrow Q$	Clause for the false rows
T	T	T	—
T	F	F	$(\sim P) \vee Q$
F	T	T	—
F	F	T	—

Only one false row, so the CNF is a single clause:

$$P \Rightarrow Q \equiv \underline{(\sim P) \vee Q}$$

That is Theorem 1.48

You proved $P \Rightarrow Q \equiv (\sim P) \vee Q$ in Week 2 by truth table, and Chartrand states it as a theorem. It is simply the CNF of the implication — and it is short because the implication is false in only one row.
Compare Wednesday: the *DNF* of the same statement had three terms, because it is true in three rows. **Whichever there are fewer of, that form is shorter.**

3. Your turn: the biconditional

P	Q	$P \Leftrightarrow Q$	Clause
T	T	T	—
T	F	F	$(\sim P) \vee Q$
F	T	F	$P \vee (\sim Q)$
F	F	T	—

$$P \Leftrightarrow Q \equiv \underline{((\sim P) \vee Q) \wedge (P \vee (\sim Q))}$$

Read the two clauses aloud: they are $P \Rightarrow Q$ and $\underline{Q \Rightarrow P}$ — which is Definition 1.52 exactly.

4. Valid, satisfiable, unsatisfiable

Three words for three situations

Term (MCS)	Chartrand's word	True in...
valid	tautology	every row
satisfiable	—	at least one row
unsatisfiable	contradiction	no row

Note that every valid formula is also satisfiable, but not conversely.

Normal forms make two questions easy

A formula is a **tautology** exactly when its CNF has no clauses — there are no false rows to build one from.
A formula is a **contradiction** exactly when its DNF has no terms.
So Monday's classification problem and today's recipe are two views of the same table.

The link between the two ideas

A formula S is satisfiable if and only if $\sim S$ is **not** valid.

5. The SAT problem

SAT

SAT is the problem: given a formula in CNF, decide whether it is satisfiable — whether *some* assignment of truth values makes it true.

Try one. Is $(P \vee Q) \wedge (\sim P \vee R) \wedge (\sim Q \vee \sim R)$ satisfiable? Find an assignment or argue none exists.

Your work

Take P true, Q false, R true. Then $(P \vee Q)$ is T, $(\sim P \vee R)$ is $F \vee T$ = T, and $(\sim Q \vee \sim R)$ is $T \vee F$ = T. **Satisfiable.**

Why this is a famous problem

The obvious method is to build the truth table and look for a T. But a formula with n variables has 2^n rows. At $n = 30$ that is over a billion; at $n = 300$ it exceeds the number of atoms in the observable universe.
Nobody knows whether there is a fundamentally faster method. **SAT was the first problem proved NP-complete**, and whether an efficient algorithm exists is the P versus NP question — the best-known open problem in computer science.

You will meet it again in theory of computation. The reason it is stated in CNF is that CNF is the standard input format for the SAT solvers used in industry, for hardware verification, scheduling, and program analysis.

The Day 1 counting argument, one last time

2^n has appeared in every unit of this course: passwords in Lecture 1, truth-table rows in Week 1, power sets next week, and now the reason SAT is hard.
"Discrete does not mean small" was the first thing we said in this course. It is also the last thing the logic unit has to say.

Exit ticket

1. How many clauses will the CNF of $P \wedge Q$ have? Explain without writing it out.

2. Is $(P \vee Q) \wedge (\sim P) \wedge (\sim Q)$ satisfiable? Justify.

Before Monday

Note: Monday, September 7 is Labor Day. Check Canvas — if there is no class, we resume Wednesday, September 9.

Read Chartrand Section 3.1, "Quantified Statements," pp. 79–84. We return to Chartrand and begin predicate logic.

Self-check (not collected). Write the CNF of $\sim P$, of $P \vee Q$, and of $P \oplus Q$, each over the variables P and Q. Check each against a truth table.

Coming up

Homework 3 is due tonight, Friday, September 4, 11:59 p.m.
Quiz 3 is Wednesday, September 9 — 15 minutes on Section 1.5 and normal forms.
Homework 4 covers Section 1.5 and normal forms, due Friday, September 11.

Questions I still have